



MATH140: Applications of Derivatives

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Learning Objectives

- Find critical points by setting $f'(x) = 0$
- Classify intervals where f is increasing or decreasing
- Apply the First and Second Derivative Tests
- Solve optimization problems using calculus

Simplify each expression completely. Show all steps and circle your final answer.

Profit maximization

1. Profit is $P(x) = -5x^2 + 16x + 94$. Find the production level x that maximizes profit and compute that maximum profit.

$$-5x^2 + 16x + 94$$

Answer: _____

2. Revenue is $R(x) = -4x^2 + 38x + 178$. Find the output level that maximizes revenue.

$$-4x^2 + 38x + 178$$

Answer: _____

3. A company's weekly profit (in thousands) is $P(x) = -2x^2 + 28x + 34$, where x is price in dollars. Find the optimal price and maximum weekly profit.

$$-2x^2 + 28x + 34$$

Answer: _____

4. Profit is $P(x) = -5x^2 + 40x + 5$. Find the production level x that maximizes profit and compute that maximum profit.

$$-5x^2 + 40x + 5$$

Answer: _____

5. Revenue is $R(x) = -2x^2 + 24x + 110$. Find the output level that maximizes revenue.

$$-2x^2 + 24x + 110$$

Answer: _____



6. A company's weekly profit (in thousands) is $P(x) = -1x^2 + 22x + 8$, where x is price in dollars. Find the optimal price and maximum weekly profit.

$$-1x^2 + 22x + 8$$

Answer: _____

7. Profit is $P(x) = -2x^2 + 32x + 46$. Find the production level x that maximizes profit and compute that maximum profit.

$$-2x^2 + 32x + 46$$

Answer: _____

8. Revenue is $R(x) = -3x^2 + 52x + 141$. Find the output level that maximizes revenue.

$$-3x^2 + 52x + 141$$

Answer: _____

9. A company's weekly profit (in thousands) is $P(x) = -2x^2 + 13x + 56$, where x is price in dollars. Find the optimal price and maximum weekly profit.

$$-2x^2 + 13x + 56$$

Answer: _____

10. Profit is $P(x) = -4x^2 + 43x + 71$. Find the production level x that maximizes profit and compute that maximum profit.

$$-4x^2 + 43x + 71$$

Answer: _____

11. Revenue is $R(x) = -4x^2 + 32x + 38$. Find the output level that maximizes revenue.

$$-4x^2 + 32x + 38$$

Answer: _____

12. A company's weekly profit (in thousands) is $P(x) = -1x^2 + 12x + 30$, where x is price in dollars. Find the optimal price and maximum weekly profit.

$$-1x^2 + 12x + 30$$

Answer: _____

13. Profit is $P(x) = -1x^2 + 27x + 26$. Find the production level x that maximizes profit and compute that maximum profit.

$$-1x^2 + 27x + 26$$

Answer: _____



14. Revenue is $R(x) = -3x^2 + 25x + 166$. Find the output level that maximizes revenue.

$$-3x^2 + 25x + 166$$

Answer: _____

15. A company's weekly profit (in thousands) is $P(x) = -2x^2 + 21x + 36$, where x is price in dollars. Find the optimal price and maximum weekly profit.

$$-2x^2 + 21x + 36$$

Answer: _____

16. Profit is $P(x) = -1x^2 + 26x + 8$. Find the production level x that maximizes profit and compute that maximum profit.

$$-1x^2 + 26x + 8$$

Answer: _____

17. Revenue is $R(x) = -1x^2 + 48x + 75$. Find the output level that maximizes revenue.

$$-1x^2 + 48x + 75$$

Answer: _____

18. A company's weekly profit (in thousands) is $P(x) = -2x^2 + 15x + 40$, where x is price in dollars. Find the optimal price and maximum weekly profit.

$$-2x^2 + 15x + 40$$

Answer: _____

19. Profit is $P(x) = -5x^2 + 34x + 55$. Find the production level x that maximizes profit and compute that maximum profit.

$$-5x^2 + 34x + 55$$

Answer: _____

20. Revenue is $R(x) = -4x^2 + 27x + 196$. Find the output level that maximizes revenue.

$$-4x^2 + 27x + 196$$

Answer: _____

21. A company's weekly profit (in thousands) is $P(x) = -1x^2 + 26x + 60$, where x is price in dollars. Find the optimal price and maximum weekly profit.

$$-1x^2 + 26x + 60$$

Answer: _____



22. Profit is $P(x) = -5x^2 + 33x + 89$. Find the production level x that maximizes profit and compute that maximum profit.

$$-5x^2 + 33x + 89$$

Answer: _____

23. Revenue is $R(x) = -2x^2 + 52x + 158$. Find the output level that maximizes revenue.

$$-2x^2 + 52x + 158$$

Answer: _____

Marginal cost and revenue

24. Total cost is $C(x) = 2x^3 + -2x^2 + 20x + 115$. Find the marginal cost $MC = C'(1)$.

$$f(x) = 2x^3 + -2x^2 + 20x + 115, x = 1$$

Answer: _____

25. Total cost is $C(x) = 1x^3 + -3x^2 + 18x + 288$. Find the marginal cost $MC = C'(1)$.

$$f(x) = 1x^3 + -3x^2 + 18x + 288, x = 1$$

Answer: _____

26. Total cost is $C(x) = 1x^3 + -2x^2 + 20x + 289$. Find the marginal cost $MC = C'(3)$.

$$f(x) = 1x^3 + -2x^2 + 20x + 289, x = 3$$

Answer: _____

27. Total cost is $C(x) = 1x^3 + -3x^2 + 5x + 107$. Find the marginal cost $MC = C'(3)$.

$$f(x) = 1x^3 + -3x^2 + 5x + 107, x = 3$$

Answer: _____

28. Total cost is $C(x) = 2x^3 + -4x^2 + 11x + 257$. Find the marginal cost $MC = C'(3)$.

$$f(x) = 2x^3 + -4x^2 + 11x + 257, x = 3$$

Answer: _____

29. Total cost is $C(x) = 2x^3 + -3x^2 + 14x + 181$. Find the marginal cost $MC = C'(2)$.

$$f(x) = 2x^3 + -3x^2 + 14x + 181, x = 2$$

Answer: _____



30. Total cost is $C(x) = 1x^3 + -5x^2 + 10x + 67$. Find the marginal cost $MC = C'(3)$.

$$f(x) = 1x^3 + -5x^2 + 10x + 67, x = 3$$

Answer: _____





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ANSWER KEY & SOLUTIONS

Topics: Marginal cost and revenue, Profit maximization. All answers verified by independent computation.

Solutions



Profit maximization

1. Profit is $P(x) = -5x^2 + 16x + 94$. Find the production level x that maximizes profit and compute that maximum profit.

$$-5x^2 + 16x + 94$$

$$\rightarrow P'(x) = -10x + 16. \text{ Set } P'(x) = 0.$$

$$\rightarrow x^* = -16/-10 = 8/5.$$

$$\rightarrow \text{Maximum profit} = P(8/5) = 534/5.$$

Answer: $x^* = -\frac{16}{-10} = \frac{8}{5}, f(x^*) = \frac{534}{5}$

2. Revenue is $R(x) = -4x^2 + 38x + 178$. Find the output level that maximizes revenue.

$$-4x^2 + 38x + 178$$

$$\rightarrow R'(x) = -8x + 38. \text{ Set equal to zero: } x = 19/4.$$

$$\rightarrow \text{Maximum revenue} = 1073/4.$$

Answer: $x^* = -\frac{38}{-8} = \frac{19}{4}, f(x^*) = \frac{1073}{4}$

3. A company's weekly profit (in thousands) is $P(x) = -2x^2 + 28x + 34$, where x is price in dollars. Find the optimal price and maximum weekly profit.

$$-2x^2 + 28x + 34$$

$$\rightarrow \text{Set } P'(x) = -4x + 28 = 0.$$

$$\rightarrow \text{Optimal price: } x = 7 \text{ dollars.}$$

$$\rightarrow \text{Maximum profit: } P(7) = 132 \text{ thousand dollars.}$$

Answer: $x^* = -\frac{28}{-4} = 7, f(x^*) = 132$

4. Profit is $P(x) = -5x^2 + 40x + 5$. Find the production level x that maximizes profit and compute that maximum profit.

$$-5x^2 + 40x + 5$$

$$\rightarrow P'(x) = -10x + 40. \text{ Set } P'(x) = 0.$$

$$\rightarrow x^* = -40/-10 = 4.$$

$$\rightarrow \text{Maximum profit} = P(4) = 85.$$

Answer: $x^* = -\frac{40}{-10} = 4, f(x^*) = 85$

5. Revenue is $R(x) = -2x^2 + 24x + 110$. Find the output level that maximizes revenue.

$$-2x^2 + 24x + 110$$

$$\rightarrow R'(x) = -4x + 24. \text{ Set equal to zero: } x = 6.$$

$$\rightarrow \text{Maximum revenue} = 182.$$

Answer: $x^* = -\frac{24}{-4} = 6, f(x^*) = 182$



6. A company's weekly profit (in thousands) is $P(x) = -1x^2 + 22x + 8$, where x is price in dollars. Find the optimal price and maximum weekly profit.

$$-1x^2 + 22x + 8$$

→ Set $P'(x) = -2x + 22 = 0$.

→ Optimal price: $x = 11$ dollars.

→ Maximum profit: $P(11) = 129$ thousand dollars.

Answer: $x^* = -\frac{22}{-2} = 11, f(x^*) = 129$

7. Profit is $P(x) = -2x^2 + 32x + 46$. Find the production level x that maximizes profit and compute that maximum profit.

$$-2x^2 + 32x + 46$$

→ $P'(x) = -4x + 32$. Set $P'(x) = 0$.

→ $x^* = -32/-4 = 8$.

→ Maximum profit = $P(8) = 174$.

Answer: $x^* = -\frac{32}{-4} = 8, f(x^*) = 174$

8. Revenue is $R(x) = -3x^2 + 52x + 141$. Find the output level that maximizes revenue.

$$-3x^2 + 52x + 141$$

→ $R'(x) = -6x + 52$. Set equal to zero: $x = 26/3$.

→ Maximum revenue = $1099/3$.

Answer: $x^* = -\frac{52}{-6} = \frac{26}{3}, f(x^*) = \frac{1099}{3}$

9. A company's weekly profit (in thousands) is $P(x) = -2x^2 + 13x + 56$, where x is price in dollars. Find the optimal price and maximum weekly profit.

$$-2x^2 + 13x + 56$$

→ Set $P'(x) = -4x + 13 = 0$.

→ Optimal price: $x = 13/4$ dollars.

→ Maximum profit: $P(13/4) = 617/8$ thousand dollars.

Answer: $x^* = -\frac{13}{-4} = \frac{13}{4}, f(x^*) = \frac{617}{8}$

10. Profit is $P(x) = -4x^2 + 43x + 71$. Find the production level x that maximizes profit and compute that maximum profit.

$$-4x^2 + 43x + 71$$

→ $P'(x) = -8x + 43$. Set $P'(x) = 0$.

→ $x^* = -43/-8 = 43/8$.

→ Maximum profit = $P(43/8) = 2985/16$.

Answer: $x^* = -\frac{43}{-8} = \frac{43}{8}, f(x^*) = \frac{2985}{16}$



11. Revenue is $R(x) = -4x^2 + 32x + 38$. Find the output level that maximizes revenue.

$$-4x^2 + 32x + 38$$

→ $R'(x) = -8x + 32$. Set equal to zero: $x = 4$.

→ Maximum revenue = 102.

Answer: $x^* = -\frac{32}{-8} = 4, f(x^*) = 102$

12. A company's weekly profit (in thousands) is $P(x) = -1x^2 + 12x + 30$, where x is price in dollars. Find the optimal price and maximum weekly profit.

$$-1x^2 + 12x + 30$$

→ Set $P'(x) = -2x + 12 = 0$.

→ Optimal price: $x = 6$ dollars.

→ Maximum profit: $P(6) = 66$ thousand dollars.

Answer: $x^* = -\frac{12}{-2} = 6, f(x^*) = 66$

13. Profit is $P(x) = -1x^2 + 27x + 26$. Find the production level x that maximizes profit and compute that maximum profit.

$$-1x^2 + 27x + 26$$

→ $P'(x) = -2x + 27$. Set $P'(x) = 0$.

→ $x^* = -27/-2 = 27/2$.

→ Maximum profit = $P(27/2) = 833/4$.

Answer: $x^* = -\frac{27}{-2} = \frac{27}{2}, f(x^*) = \frac{833}{4}$

14. Revenue is $R(x) = -3x^2 + 25x + 166$. Find the output level that maximizes revenue.

$$-3x^2 + 25x + 166$$

→ $R'(x) = -6x + 25$. Set equal to zero: $x = 25/6$.

→ Maximum revenue = $2617/12$.

Answer: $x^* = -\frac{25}{-6} = \frac{25}{6}, f(x^*) = \frac{2617}{12}$

15. A company's weekly profit (in thousands) is $P(x) = -2x^2 + 21x + 36$, where x is price in dollars. Find the optimal price and maximum weekly profit.

$$-2x^2 + 21x + 36$$

→ Set $P'(x) = -4x + 21 = 0$.

→ Optimal price: $x = 21/4$ dollars.

→ Maximum profit: $P(21/4) = 729/8$ thousand dollars.

Answer: $x^* = -\frac{21}{-4} = \frac{21}{4}, f(x^*) = \frac{729}{8}$



16. Profit is $P(x) = -1x^2 + 26x + 8$. Find the production level x that maximizes profit and compute that maximum profit.

$$-1x^2 + 26x + 8$$

$$\rightarrow P'(x) = -2x + 26. \text{ Set } P'(x) = 0.$$

$$\rightarrow x^* = -26/-2 = 13.$$

$$\rightarrow \text{Maximum profit} = P(13) = 177.$$

Answer: $x^* = -\frac{26}{-2} = 13, f(x^*) = 177$

17. Revenue is $R(x) = -1x^2 + 48x + 75$. Find the output level that maximizes revenue.

$$-1x^2 + 48x + 75$$

$$\rightarrow R'(x) = -2x + 48. \text{ Set equal to zero: } x = 24.$$

$$\rightarrow \text{Maximum revenue} = 651.$$

Answer: $x^* = -\frac{48}{-2} = 24, f(x^*) = 651$

18. A company's weekly profit (in thousands) is $P(x) = -2x^2 + 15x + 40$, where x is price in dollars. Find the optimal price and maximum weekly profit.

$$-2x^2 + 15x + 40$$

$$\rightarrow \text{Set } P'(x) = -4x + 15 = 0.$$

$$\rightarrow \text{Optimal price: } x = 15/4 \text{ dollars.}$$

$$\rightarrow \text{Maximum profit: } P(15/4) = 545/8 \text{ thousand dollars.}$$

Answer: $x^* = -\frac{15}{-4} = \frac{15}{4}, f(x^*) = \frac{545}{8}$

19. Profit is $P(x) = -5x^2 + 34x + 55$. Find the production level x that maximizes profit and compute that maximum profit.

$$-5x^2 + 34x + 55$$

$$\rightarrow P'(x) = -10x + 34. \text{ Set } P'(x) = 0.$$

$$\rightarrow x^* = -34/-10 = 17/5.$$

$$\rightarrow \text{Maximum profit} = P(17/5) = 564/5.$$

Answer: $x^* = -\frac{34}{-10} = \frac{17}{5}, f(x^*) = \frac{564}{5}$

20. Revenue is $R(x) = -4x^2 + 27x + 196$. Find the output level that maximizes revenue.

$$-4x^2 + 27x + 196$$

$$\rightarrow R'(x) = -8x + 27. \text{ Set equal to zero: } x = 27/8.$$

$$\rightarrow \text{Maximum revenue} = 3865/16.$$

Answer: $x^* = -\frac{27}{-8} = \frac{27}{8}, f(x^*) = \frac{3865}{16}$



21. A company's weekly profit (in thousands) is $P(x) = -1x^2 + 26x + 60$, where x is price in dollars. Find the optimal price and maximum weekly profit.

$$-1x^2 + 26x + 60$$

→ Set $P'(x) = -2x + 26 = 0$.

→ Optimal price: $x = 13$ dollars.

→ Maximum profit: $P(13) = 229$ thousand dollars.

Answer: $x^* = -\frac{26}{-2} = 13, f(x^*) = 229$

22. Profit is $P(x) = -5x^2 + 33x + 89$. Find the production level x that maximizes profit and compute that maximum profit.

$$-5x^2 + 33x + 89$$

→ $P'(x) = -10x + 33$. Set $P'(x) = 0$.

→ $x^* = -33/-10 = 33/10$.

→ Maximum profit = $P(33/10) = 2869/20$.

Answer: $x^* = -\frac{33}{-10} = \frac{33}{10}, f(x^*) = \frac{2869}{20}$

23. Revenue is $R(x) = -2x^2 + 52x + 158$. Find the output level that maximizes revenue.

$$-2x^2 + 52x + 158$$

→ $R'(x) = -4x + 52$. Set equal to zero: $x = 13$.

→ Maximum revenue = 496.

Answer: $x^* = -\frac{52}{-4} = 13, f(x^*) = 496$



Marginal cost and revenue

24. Total cost is $C(x) = 2x^3 + -2x^2 + 20x + 115$. Find the marginal cost $MC = C'(1)$.

$$f(x) = 2x^3 + -2x^2 + 20x + 115, x = 1$$

$$\rightarrow MC = C'(x) = 6x^2 + -4x + 20.$$

$$\rightarrow MC \text{ at } x = 1: C'(1) = 22.$$

Answer: $f'(x) = 6x^2 + -4x + 20, f'(1) = 22$

25. Total cost is $C(x) = 1x^3 + -3x^2 + 18x + 288$. Find the marginal cost $MC = C'(1)$.

$$f(x) = 1x^3 + -3x^2 + 18x + 288, x = 1$$

$$\rightarrow MC = C'(x) = 3x^2 + -6x + 18.$$

$$\rightarrow MC \text{ at } x = 1: C'(1) = 15.$$

Answer: $f'(x) = 3x^2 + -6x + 18, f'(1) = 15$

26. Total cost is $C(x) = 1x^3 + -2x^2 + 20x + 289$. Find the marginal cost $MC = C'(3)$.

$$f(x) = 1x^3 + -2x^2 + 20x + 289, x = 3$$

$$\rightarrow MC = C'(x) = 3x^2 + -4x + 20.$$

$$\rightarrow MC \text{ at } x = 3: C'(3) = 35.$$

Answer: $f'(x) = 3x^2 + -4x + 20, f'(3) = 35$

27. Total cost is $C(x) = 1x^3 + -3x^2 + 5x + 107$. Find the marginal cost $MC = C'(3)$.

$$f(x) = 1x^3 + -3x^2 + 5x + 107, x = 3$$

$$\rightarrow MC = C'(x) = 3x^2 + -6x + 5.$$

$$\rightarrow MC \text{ at } x = 3: C'(3) = 14.$$

Answer: $f'(x) = 3x^2 + -6x + 5, f'(3) = 14$

28. Total cost is $C(x) = 2x^3 + -4x^2 + 11x + 257$. Find the marginal cost $MC = C'(3)$.

$$f(x) = 2x^3 + -4x^2 + 11x + 257, x = 3$$

$$\rightarrow MC = C'(x) = 6x^2 + -8x + 11.$$

$$\rightarrow MC \text{ at } x = 3: C'(3) = 41.$$

Answer: $f'(x) = 6x^2 + -8x + 11, f'(3) = 41$

29. Total cost is $C(x) = 2x^3 + -3x^2 + 14x + 181$. Find the marginal cost $MC = C'(2)$.

$$f(x) = 2x^3 + -3x^2 + 14x + 181, x = 2$$

$$\rightarrow MC = C'(x) = 6x^2 + -6x + 14.$$

$$\rightarrow MC \text{ at } x = 2: C'(2) = 26.$$

Answer: $f'(x) = 6x^2 + -6x + 14, f'(2) = 26$



30. Total cost is $C(x) = 1x^3 + -5x^2 + 10x + 67$. Find the marginal cost $MC = C'(3)$.

$$f(x) = 1x^3 + -5x^2 + 10x + 67, \quad x = 3$$

$$\rightarrow MC = C'(x) = 3x^2 + -10x + 10.$$

$$\rightarrow MC \text{ at } x = 3: C'(3) = 7.$$

Answer: $f'(x) = 3x^2 + -10x + 10, \quad f'(3) = 7$

